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Ultrastable long-distance fibre-optic time transfer: active compensation over a wide range of delays

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Abstract

In this paper we describe a new solution of active delay stabilization for fibre-optic distribution of time and RF-frequency signals, which allows one to obtain both high precision and a potentially unlimited range of compensation of the fibre delay fluctuations. The solution is based on a hybrid system exploiting a pair of continuously tuned electronic variable delay lines, and a set of switched optical delays. We present a fully operational prototype of the time and frequency distribution setup based on this idea, which is capable of compensating more than 1 µs of the fiber delay fluctuations, and thus may be used in very long-haul links up to about 1000 km, without the need for any seasonal maintenance. We also report measurements of the time and frequency distribution stability, and the verification of the time transfer calibration.

Keywords: time transfer, frequency transfer, fibre optics

(Some figures may appear in colour only in the online journal)

1. Introduction

In recent years, many solutions for fibre-optic time and frequency (T & F) transfer have been proposed. To obtain high transfer stability it is necessary to undertake some means to reduce the impact of delay fluctuations, observed in fibres due to temperature variations. Generally, there are two groups of solutions: the first one is a two-way transfer over optical fibre [1–5] similar in principle to the two-way satellite time and frequency transfer (TWSTFT) method, and the second one uses various ideas for compensation of these delay fluctuations [6–13]. The serious advantage of the second group of methods is that they are suitable not only for classical comparison of two distant clocks or timescale realizations, but may also be used for T & F distribution.

In previous papers, we described the system for T & F distribution, based on active stabilization of the link delay by means of continuously tuned electronic delay lines [12, 13]. The delay lines designed as an application-specific integrated circuit (ASIC) allow one to compensate up to 100 ns of delay fluctuations, which is sufficient for compensating annual delay drifts of about 100 km of soil-buried fibre. In very long haul links however, which may cover distances of many hundreds of kilometres [4, 14–16], this compensation range was insufficient. Thus, there was a need for the occasional addition or removal of some extra optical delays (i.e. short sections of fibre), which causes a short interruption in signal transmission and engages human activity. Such a situation occurs in our first long-haul installation, connecting two Polish UTC laboratories: UTC(PL) and UTC(AOS) over a 420 km-long fibre [17, 18], where maintenance activity is needed a few times during the year—see figure 1(*a*).

In this paper, we describe an idea of a fully automated extension of the compensation range to more than 1 µs, allowing continuous, maintenance-free operation of the system over up to 1000 km of fibre.

2. Hybrid compensation of delay fluctuations

A block diagram of our improved T & F distribution system is depicted in figure 2(*a*), with the added blocks and signal

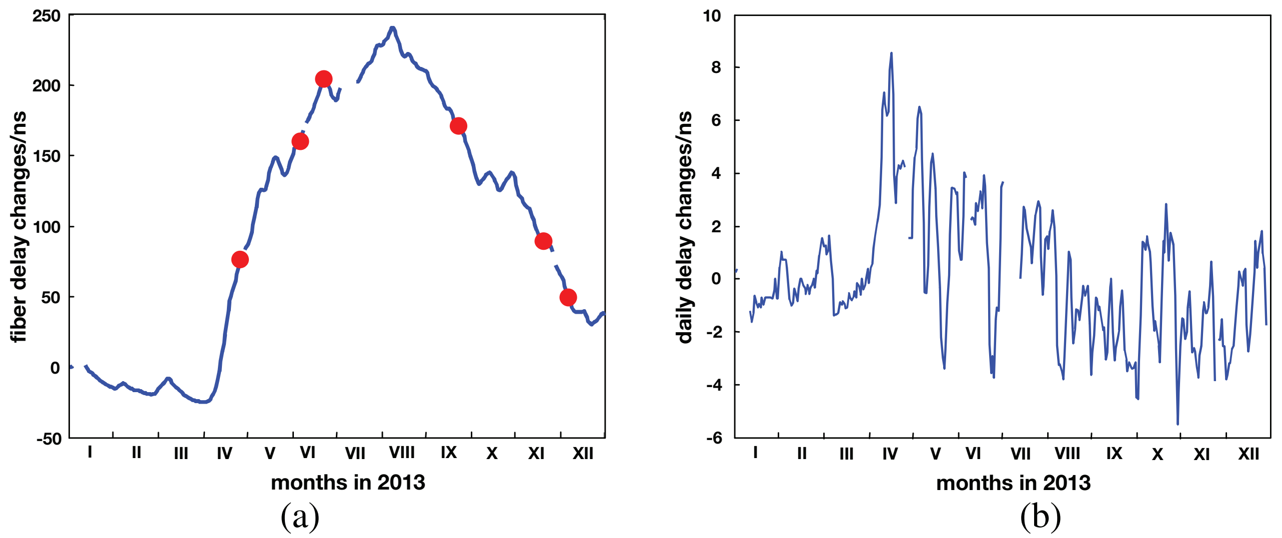

Figure 1. Measured annual (*a*) and daily (*b*) delay fluctuations in 420 km-long link Warszawa–Borowiec. The dots in (*a*) mark the manual adding/removing of additional optical delays. (Both plots are based on a one-per-day registration of the delay fluctuations. The gaps in the plots are due to a lack of data for particular periods.)

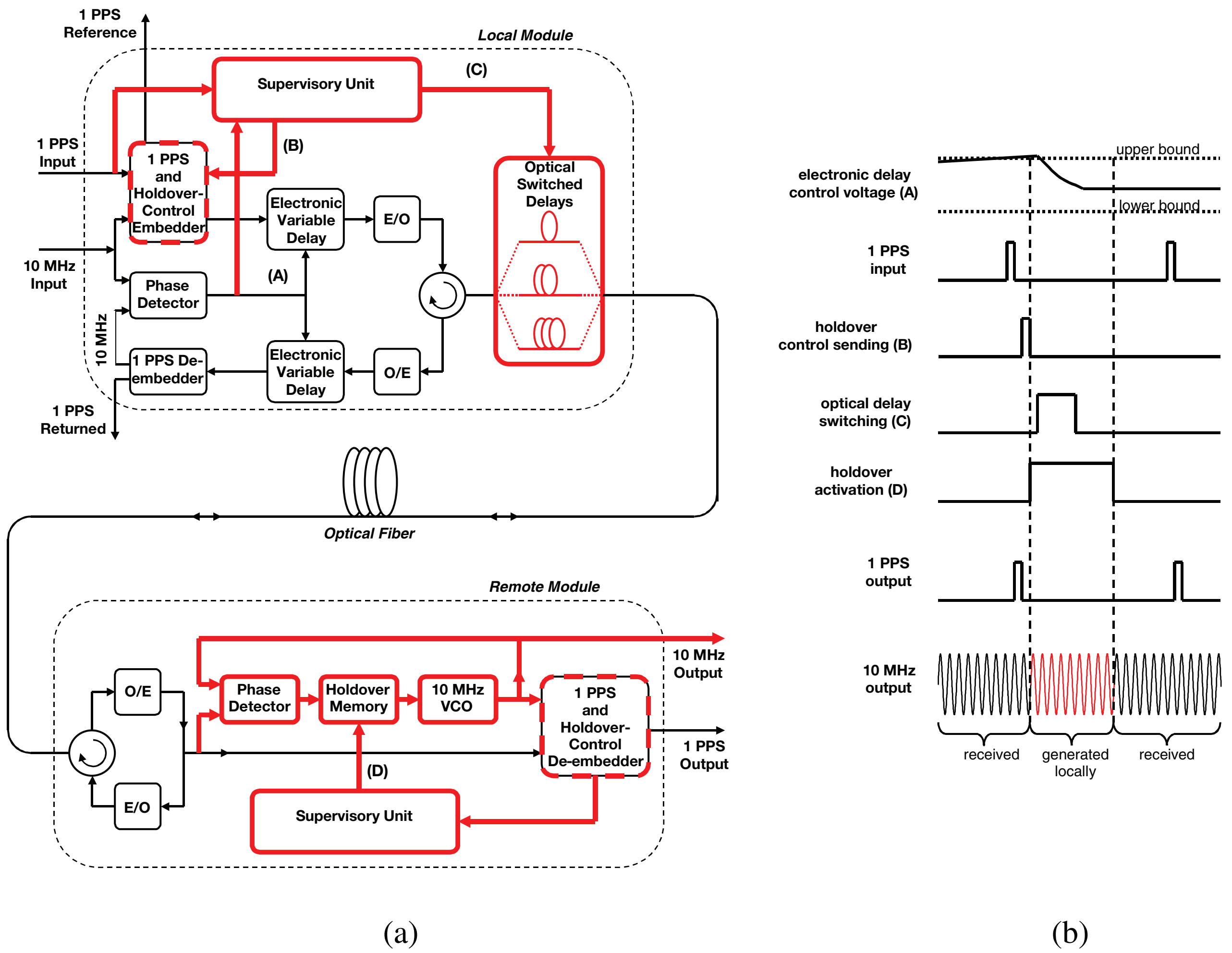

Figure 2. Block diagram of a T & F distribution system with an extended delay compensation range (*a*), and a schematic of the event sequence during optical delay switching (*b*).

paths marked by bold lines and the modified blocks marked by dashed bold lines. Recalling the principal description of our basic system, presented in extension in [12, 13], it should be noted that the active delay stabilization is obtained in the feedback system, which redirected the signal reaching the remote module backwards to the local module, where the

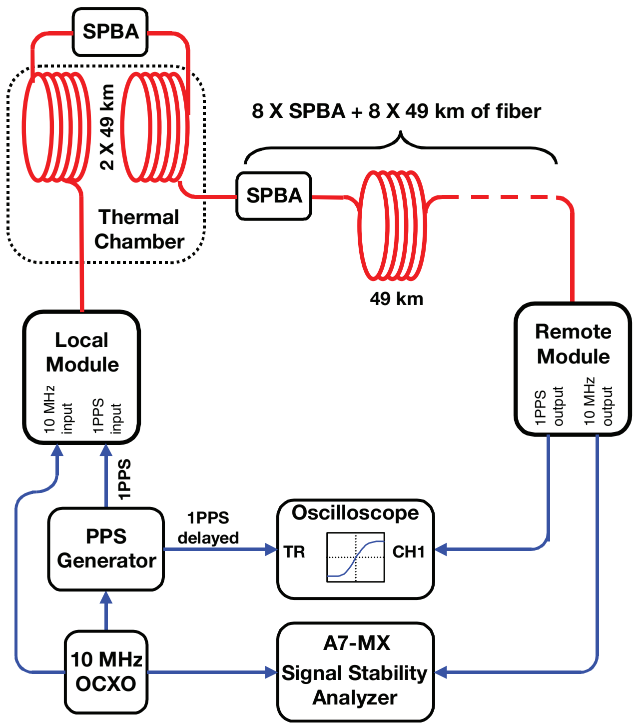

Figure 3. Basic measurement setup.

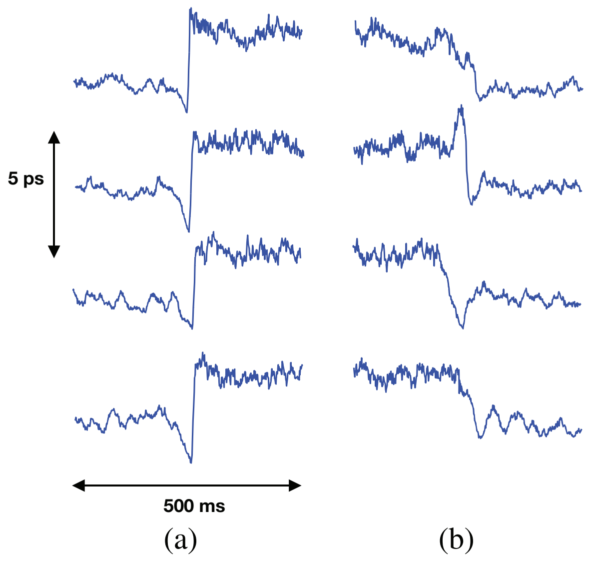

Figure 4. Eight examples of phase excursions during the optical delay switching. Column (*a*) illustrates the case of switching resulting in increased optical delay with simultaneously decreased electronic delay, and column (*b*), the opposite case.

phase detector controls the two precisely matched electronic delay lines to compensate the delay fluctuations occurring in the fibre. The tuning range of this delay line is 100 ns, and it seems to be rather impossible to extend the compensation range without noticeable degradation of the system performance [19]. Thus, we extended the compensation range by adding additional switched optical delays in series with the fibre link, which would eliminate the need for manual operations, currently performed in our first long-haul system. The main challenge in this new solution was to ensure continuous system operation, without any considerable signs of switching and relocking being observed in the output signals. The main idea is to send to the remote module the forestall information about the impending switching event, which triggers off a holdover mode in the remote module for a short period needed for switching and relocking of the feedback system. In holdover mode the output frequency signal is temporarily generated locally by a highly stable oscillator, 'covering' the transmission discontinuity.

In the local module, the additional blocks are a bank of switched optical delays and the supervisory unit. The bank of optical delays consists of six optical switches and five sections of fibre (10, 20, 40, and two times 80 m), which allows for the introduction of a delay varied from (nearly) zero to 1150 ns, in 23 steps, 50 ns each. (The insertion loss of the optical delays is 2.5 dB.) Also the 1 PPS embedder is redesigned to interweave to the outgoing signal new information about the switching action, which starts the holdover mode in the remote module. In the remote module, the 1 PPS deembedder is fitted up with a holdover-control signal detector. An additional 10 MHz voltage control oscillator (VCO) with a phase detector and holdover memory constitute a phase-locked loop (PLL), which may be driven to holdover by 'freezing' the VCO control voltage in a holdover memory. The supervisory unit in the remote module determines the proper duration of the holdover mode.

All the action starts when a voltage controlling electronic delay line approaches the upper or lower bound, concerned with the end of the ASIC's tuning range (see figure 2(*b*)). Next, just after 1 PPS input pulse, the holdover signal is sent to the remote module, where it activates the holdover mode. After some small latency, the optical delay is switched, and then the equilibrium state of the feedback delay-stabilizing loop is recovered. It should be stressed that the system input-to-output delay before and after switching is the same, because the change of optical delay results in an opposite change of electronic variable delays, as during the normal system operation. After about 500 ms all the transient processes are finished, and the whole system returns to normal operation well before the next 1 PPS pulse. Thus the time transfer is unaffected by the switching process.

For the 10 MHz signal however, the holdover mode breaks the normal operation of the PLL in the remote module, which means that in this period the VCO is not locked to the frequency reference received via the fibre, but goes in an open loop. The short-term stability of this oscillator is thus essential for a smooth 10 MHz generation in the holdover period, with a possible low phase drift.

3. Measurement results

The experimental setup we used to check the performance of the proposed solution is shown in figure 3. The local and remote modules of our system were connected with a 490 km-long optical path containing ten spools of standard SMF-28 fibre (each of 49 km) and nine single-path bidirectional optical amplifiers (SPBAs). Two spools were placed in a thermal chamber with a heating/cooling period of one day, and the

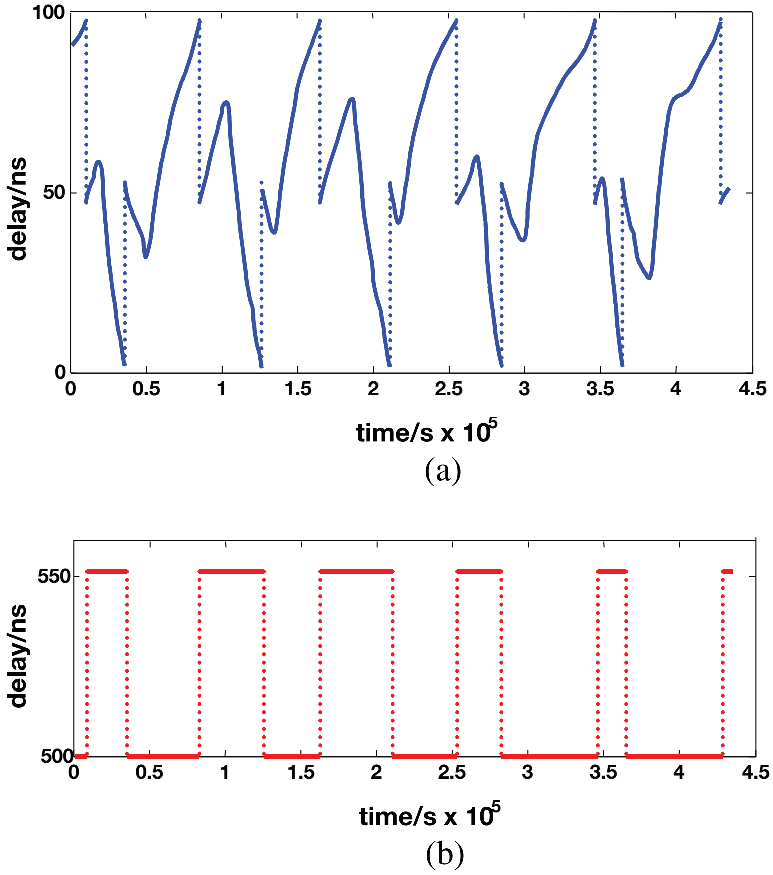

Figure 5. Registered changes of electronic (*a*) and optical (*b*) delays caused by daily heating/cooling cycles applied to the fibre.

remaining eight spools and all the amplifiers were located in a laboratory not equipped with air conditioning. The rest of the equipment was in a temperature-stabilized room, to reduce the impact of temperature on the measurement equipment. The phase stability of the 10 MHz output signal was measured with an A7-MX signal stability analyser, and 1 PPS delay stability was analysed using a high-speed digital oscilloscope (DSO81004 type). The oscilloscope was triggered by the 1 PPS signal delayed relative to the signal entering the local module, which allows the time interval measured by the oscilloscope to be reduced to no greater than a few microseconds.

In the first experiment, we examined the stability of the transmitted 10 MHz signal in a case of optical delay switching. During these measurements, we did not change the temperature in the chamber, but we provoked frequent switching events using the software controlling the local module. The phase of the 10 MHz output signal was registered using A7-MX with a sampling rate of 1 ms. Two hundred switching events were registered and analysed, and eight typical examples are shown in figure 4. One may notice some transient phase excursion as well as a small, steady phase shift concerned with the switching. The transient phase excursions during holdover are concerned with a phase drift of temporarily uncontrolled VCO, whereas the steady phase shift is caused by a residual mismatch of the two electronic delay lines, in which the operating point changes substantially during the switching. All these effects are below 5 ps.

In the main experiment, we applied daily heating/cooling cycles with a magnitude of 25 °C to the spools in the chamber, together with natural temperature fluctuations of about 5 °C affecting the remaining eight spools. The sampling period of the measurement setup, the same for both 10 MHz and 1 PPS signals, was 10 s.

As the fibre delay variations exceeded the tuning range of the electronic delay lines, the switching of optical delays occurred twice each day. The experiment lasted for five days. It should be pointed out that this test was intentionally extremely stressful—in our real 420 km-long link the typical daily fibre delay changes are no greater than a few nanoseconds (see figure 1(*b*)), and thus in a normal situation switching would occur no more than once every few weeks.

The registered changes of both the electronic and optical delays are presented in figure 5. One may notice that when the electronic delays approach their limits, the optical delay is changed by about 50 ns, and thus the electronic delays return to the middle of their tuning range.

Figure 6 shows the residual phase fluctuations of the 10 MHz signal and the resulting overlapping Allan deviation (ADEV). The observed phase excursions were about 12 ps peak–peak, and the ADEV slope was close to τ^{-1} over the entire observation period, with a slight bump located in the region of 4×10^4 s, which is consistent with the daily cycles of fibre heating. When the average time equals 10^5 s the ADEV is below 3×10^{-17}. In figure 6(*b*) we also included an ADEV

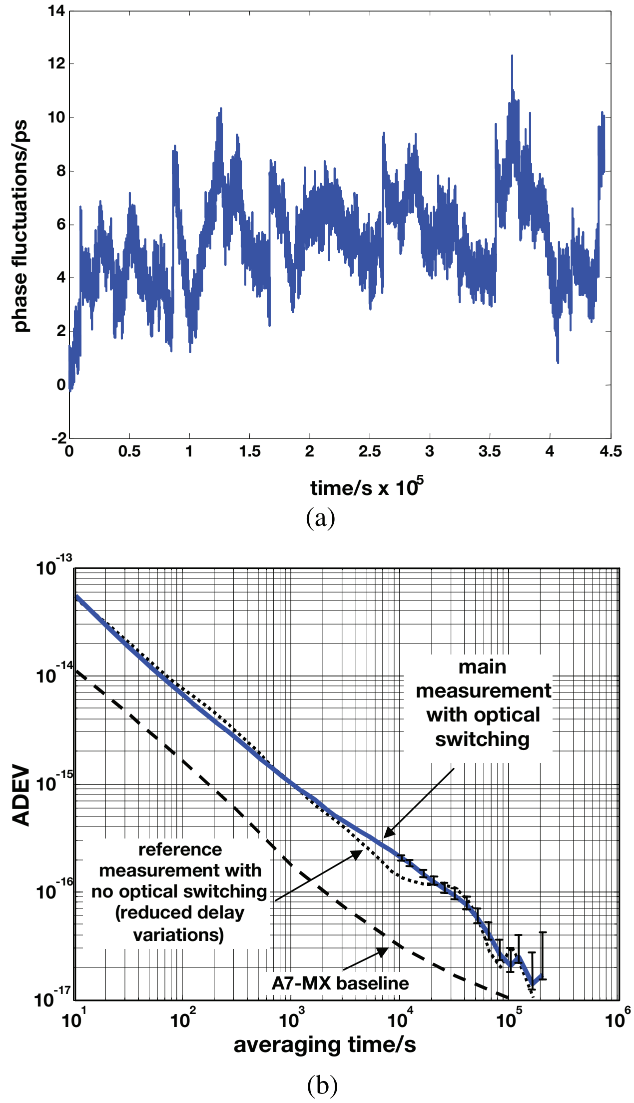

Figure 6. 10 MHz output stability presented in the form of time-domain phase excursions (*a*) and overlapping ADEV (*b*).

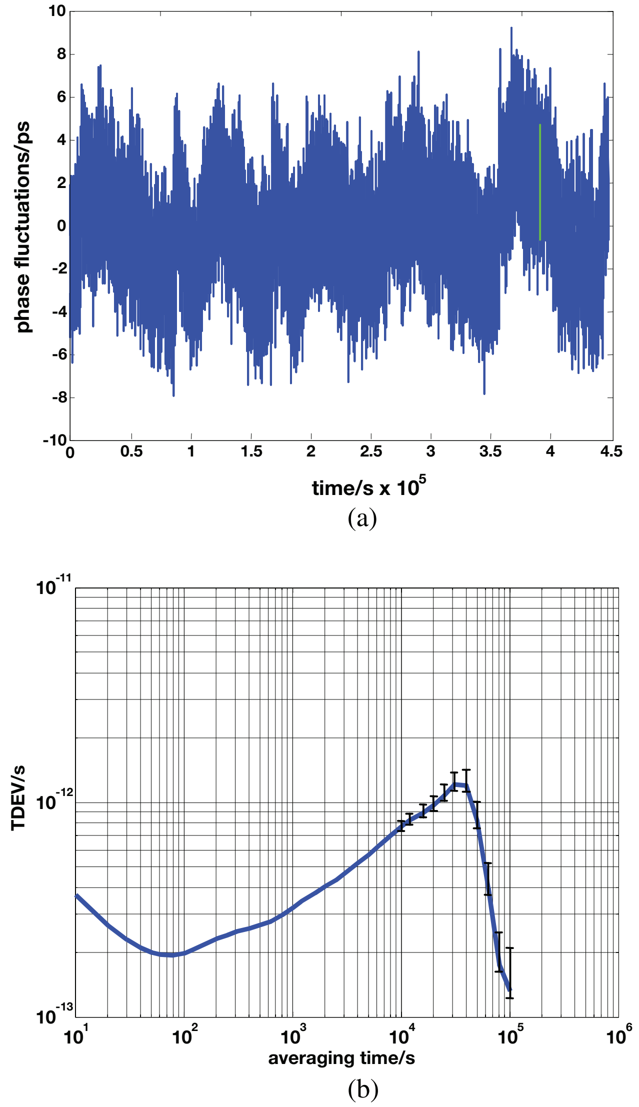

Figure 7. 1 PPS output stability presented in the form of time-domain phase excursions (*a*), and TDEV (*b*).

plot obtained for reduced temperature variations, for which the delay changes were about 18 ns, and thus there were no switching processes in the system. One may notice that there is no significant impact of the optical switching on the output signal stability. (The characteristic of the half-a-day bump that is more spread in the case of switching is probably due to some irregularity of the switching events—compare figure 5(*b*).)

The stability of the 1 PPS signal register during the same experiment is shown in figure 7. The temporal evolution of the 1 PPS phase is closed to that of the 10 MHz signal (compare figures 6(*a*) and 7(*a*)), but is noticeably more noisy, because of the noticeable impact of the timebase jitter of the oscilloscope used in the measurement. The time deviation (TDEV) slightly exceeds 1 ps for half-a-day averaging, and is significantly lower for both the shorter and longer averaging time.

Additionally we verified a procedure of time transfer calibration, which we developed and examined earlier in regard to the basic system without optical switches. Generally, the calibration of a local-to-remote delay of the 1 PPS signal is performed by measuring the round-trip delay, dividing this value by two, and applying some corrections, concerned with fibre chromatic dispersion, the Sagnac effect, and internal delays of the hardware. (The procedure is thoroughly described in [13, 17], together with the uncertainty analysis.) Basically, the calibration algorithm for the new system is exactly the same, and we performed the verification only because of a redesign of some crucial pieces of hardware. In the experiment we used the same method and equipment as described previously, thus in this work we present only brief results. We examined the calibration using 98 km, 196 km, 294 km, 392 km and 490 km of spooled fibre of the SMF-28 type. Both the local and remote modules were on the same lab bench, so that we could perform both the calibration procedure, and direct measurement of the delay between the reference point in the local module and the output of the remote module, to evaluate the calibration accuracy. The results are presented in table 1. One may notice that the results of the calibration are consistent with the direct measurements, with the differences not exceeding the estimated standard uncertainty of the results.

Table 1. Comparison of the 1 PPS signal delay obtained from the calibration procedure and measured directly. The estimated standard uncertainty is in brackets.

Fibre length/km	98	196	294	392	490
Delay obtained from calibration/ps	497 247 489 (±9)	992 248 160 (±12)	1 487 248 833 (±15)	1 983 249 501 (±19)	2 478 750 198 (±24)
Delay measured directly/ps	497 247 491 (±5)	992 248 169 (±5)	1 487 248 844 (±5)	1 983 249 500 (±5)	2 478 750 201 (±5)
Difference/ps	2 (±10)	9 (±13)	11 (±16)	−1 (±20)	3 (±26)

4. Conclusion

In this paper we presented a new solution of actively stabilized time and frequency transfer with a hybrid compensation of the fibre delay fluctuations. The continuous (analogue) compensation is performed by electronic tunable delay lines, and it is occasionally supplemented by the switching of optical delays. The most challenging problem was to obtain continuous delivery of time and frequency signals to the outputs of the remote module, without having the quality deteriorated by the switching process. Evaluating the fully operational prototype of the system we demonstrated experimentally that during the switching process, phase excursions of the 10 MHz output signal were below 5 ps, and the1 PPS transmission was unaffected, as the switching process was located between two consecutive 1 PPS pulses. In a stressful test, when the switching occurred twice a day, we measured the ADEV of the 10 MHz signal and the TDEV of the 1 PPS signal, and we found the stability parameters to be only slightly worse compared to earlier measurements, where the fibre delay changes were substantially lower.

At the end of the year 2014 we are planning to install the system presented herein between the Astrogeodynamic Observatory (AOS) in Borowiec near Poznań and The National Laboratory for Atomic Molecular and Optical Physics (FAMO) in Toruń, over about 350 km of fibre. In 2015 our first link between UTC(PL) in Warsaw and UTC(AOS) in Borowiec will also be upgraded with this new solution, which will give an opportunity for further, long term evaluation of the presented system in a real environment.

Acknowledgments

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References

[1] Rost M, Piester D, Yang W, Feldmann T, Wubbena T and Bauch A 2012 Time transfer through optical fibres over a distance of 73 km with an uncertainty below 100 ps *Metrologia* **49** 772–82

[2] Lopez O, Kanj A, Pottie P E, Rovera D, Achkar J, Chardonnet C, Amy-Klein A and Santarelli G 2012 Simultaneous remote transfer of accurate timing and optical frequency over a public fiber network *Appl. Phys.* **110** 3–6

[3] Smotlacha V and Kuna A 2012 Two-way optical time and frequency transfer between IPE and BEV *Proc. European Frequency and Time Forum 2012* (*Gothenburg, Sweden, April 2012*)

[4] Amemiya M, Imae M, Fujii Y, Suzuyama T, Hong F and Takamoto M 2010 Precise frequency comparison system using bidirectional optical amplifiers *IEEE Trans. Instrum. Meas.* **59** 632–40

[5] Calosso C E, Bertacco E, Calonico D, Clivati C, Costanzo G A, Frittelli, M, Levi F, Mura A and Godone A 2014 Frequency transfer via a two-way optical phase comparison on a multiplexed fiber network *Opt. Lett.* **39** 1177–80

[6] Lopez O, Amy-Klein A, Lours M, Chardonnet C and Santarelli G 2010 High-resolution microwave frequency dissemination on an 86-km urban optical link *Appl. Phys.* B **98** 723–7

[7] Ebenhag S C, Wingqvist C and Hedekvist P O 2012 Real time phase stable one way frequency transfer over optical fiber *Proc. European Frequency and Time Forum 2012* (*Gothenburg, Sweden, 24–26 April, 2012*)

[8] Hanssen J, Crane S and Ekstrom C 2012 One-way two-color fiber link for frequency transfer *Proc. 2012 IEEE Int. Frequency Control Symp.* (*Baltimore, MD, 21-24 May 2012*) pp 1–4

[9] Marra G, Margolis H S and Richardson D J 2012 Dissemination of an optical frequency comb over fiber with 3×10^{-18} fractional accuracy *Opt. Express* **20** 1775–82

[10] Wang B, Gao C, Chen W L, Miao J, Zhu X, Bai Y, Zhang J W, Feng Y Y, Li T C and Wang L J 2012 Precise and continuous time and frequency synchronisation at the 5×10^{-19} accuracy level *Sci. Rep.* **2** 556

[11] He Y, Orr B J, Baldwin K G H, Wouters M J, Luiten A N, Aben G and Warrington R B 2013 Stable radio-frequency transfer over optical fiber by phase-conjugate frequency mixing *Opt. Express* **21** 18754–64

[12] Śliwczyński Ł, Krehlik P, Buczek Ł and Lipiński M 2011 Active propagation delay stabilization for fiber optic frequency distribution using controlled electronic delay lines *IEEE Trans. Instrum. Meas.* **60** 1480–8

[13] Krehlik P, Śliwczyński Ł, Buczek Ł and Lipiński M 2012 Fiber optic joint time and frequency transfer with active stabilization of the propagation delay *IEEE Trans. Instrum. Meas.* **61** 2844–51

[14] Predehl K *et al* 2012 A 920 k optical fiber link for frequency metrology at the 19th decimal place *Science* **336** 441–4

[15] Śliwczyński Ł, Krehlik P, Buczek Ł and Lipiński M 2012 Frequency transfer in electronically stabilized fiber optic link exploiting bidirectional optical amplifiers *IEEE Trans. Instrum. Meas.* **61** 2573–80

[16] Śliwczyński Ł and Kołodziej J 2012 Bidirectional optical amplification in long-distance two-way fiber-optic time and frequency transfer systems *IEEE Trans. Instrum. Meas.* **62** 253–62

[17] Śliwczyński Ł, Krehlik P, Czubla A, Buczek Ł and Lipiński M 2013 Dissemination of time and RF frequency via a stabilized fibre optic link over a distance of 420 km *Metrologia* **50** 133–45

[18] Jiang Z, Czubla A, Nawrocki J, Lewandowski W and Arias F 2014 Towards accurate optical fibre time transfer in UTC *Proc. European Frequency and Time Forum 2014 (Neuchatel, 23–26 June 2014)* pp 231–4

[19] Krehlik P, Śliwczyński Ł and Idzik M 2014 Extending the delay compensation range of the fiber optic time and frequency transfer system *Proc. European Frequency and Time Forum 2014 (Neuchatel, 23–26 June 2014)* pp 346–8